Homework 4

BIOSTAT 740, Fall 2026

Due October 9, 8:00 PM

Purpose:

1. Choose a covariance structure

HW4 tasks

1. Use SAS and the exercise data for the following
2. Use **only** times 0,4,8, and 12, and fit model with two different lines for the mean and choose the best among the following covariance structures:
    a. Compound symmetry
    b. Heterogenous compound symmetry
    c. Exponential
    d. Heterogeneous autoregressive
    e. Unstructured
3. Repeat the exercise from part 2, but using time points 0, 4, 6, 8, 10, and 12.
4. Did you get the same answer from part 2 and part 3? Why do you think that might be?

Coding hints

1. You will have to be careful about missing data. This data set has both drop-outs and intermittent missingness– make sure your software is handling that correctly.
2. You will likely have to extract the numeric part of a character (string) variable during a data step. I do this using the `substr` function. For example: `y = substr(_name_,2);`.
3. You will also likely have to convert a string variable into a numeric variable. One quick way to do this in SAS is to perform some numeric function on it. I often will multiply it by 1.0. You could also add 0 to it.

What to turn in

Please submit **on Canvas** a PDF files with

1. Annotated SAS code with your selected structures and discussion for part 4.

Cutting and pasting is fine! For your SAS result, it is fine to cut and paste results into your .SAS file, type in your conclusions and thoughts and then save it as a PDF.